Diabetes and Insulin Injections Measured Through Calculus

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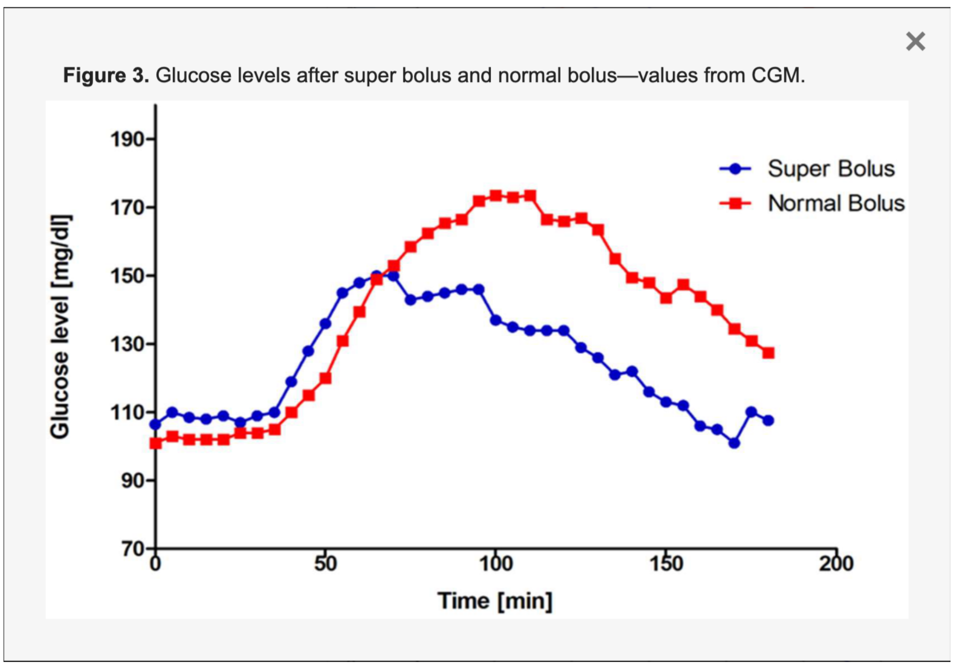
Diabetes is a disease that occurs in over 800 million people worldwide and happens when a person's blood glucose levels are too high. The hormone insulin is the primary concern, as in Type 1 Diabetes, the cells that make insulin are destroyed by the immune system, and in Type 2 Diabetes, the insulin target cells do not respond to insulin normally, causing resistance. Thus, insulin injections are necessary for people with diabetes, but these injections must be given at the correct time to have the most effective results.

As a pre-health major, discovering how different bodies work and what modern medicine can do to combat these abnormal bodily processes is interesting to me. I enjoy exploring the how and why behind the what, as it helps me to better understand what one can do moving forward to be healthy and feel their best. Diabetes is a disease that runs in my family, so being able to understand what can be done to assist those with diabetes is not only fascinating, but rewarding as well. Knowing when to inject insulin into people with lack of or resistance to insulin is important in the field of endocrinology because the hormone insulin is the one responsible for lowering blood glucose levels and helps the body maintain normal blood glucose levels. Diabetes can cause complications such as damage to the blood vessels, kidney problems, and nerve issues, so maintaining normal glucose levels is vital. In this paper, my objective is to determine the appropriate time to inject insulin so that when glucose spikes or is at its maximum point, insulin is working most efficiently.

I apply calculus to a graph that compares blood glucose levels prior to a meal and immediately after a Super Bolus insulin injection and a Normal Bolus insulin injection by finding a function that appropriately reads the graph. The Super Bolus is used for a high glycemic index food intake, while a normal bolus is used when a normal amount of carbs is ingested. To find the highest spike in glucose levels, I apply the derivative to the function and set it equal to zero to find the critical point. Then, to find when the blood glucose levels are dropping at the quickest rate—or when insulin is most effective on this subject—I find the inflection point of the derivative by taking the derivative of the first derivative, otherwise known as the second derivative. Finding this rate from this graph is important because we can use it to determine how far in advance insulin must be injected by subtracting the time of the graph at the peak glucose level from the time glucose is dropping the fastest.

Calculations:



First, we must determine the function of the graph.

Glucose concentration curve function:

$$G(t) = A \cdot t \cdot e^{-kt}$$

Parameters:

$G(t)$ = glucose concentration in mg/dL (milligrams per deciliters) at time t in minutes

A = amplitude factor, dependent on the k value, and provides accuracy for the model to the real-world example

t = time in minutes

e = natural logarithm function to express exponential growth

k = rate constant per minute

To find the peak of the curve, we must find the derivative of the function. Separating the function into two different functions $A \cdot t$ and e^{-kt}, we can find the derivative of each function separately, using the product rule in the first and chain rule in the second, then use the product rule to get G'(t).

$$G'(t) = ((0)(t) + (1)(A)) \cdot (-ke^{-kt})$$

$$G'(t) = (A)(e^{-kt}) + (At)(-ke^{-kt})$$

$$G'(t) = Ae^{-kt} - Akte^{-kt}$$

$$G'(t) = Ae^{-kt}(1 - kt)$$

We can use the derivative to find when the glucose concentration is at its maximum by setting the equation to zero:

$$0 = Ae^{-kt}(1 - kt)$$

A cannot be 0 because it would mean the blood glucose level never increased, and e^{-kt} cannot be 0 because e^x must always be positive. That leaves the equation to be

$$0 = 1 - kt$$

Then we can add 1 to both sides and divide by k to find the value of t.

$$t = 1 / k$$

By looking at the graph and focusing on the Super Bolus curve, the peak is reached when $t = 100$ minutes. This allows us to find our k value, or our constant, through the equation 1/100 minutes, which is equal to .0100 per minute.

Using this information, we can find the second derivative. The second derivative tells us the concavity of the original graph and where the derivative reaches its minimum value; further, this tells us when glucose concentration drops the quickest. This is important so we can determine when to inject insulin so it aligns with when glucose is at its peak. We can split up the equation into two functions again, using the chain rule for the first and the power rule for the second.

$$Ae^{-kt} = Ae^{-kt} \cdot -k$$

$$= -kAe^{-kt}$$

$$1 - kt = -k$$

$$G''(t) = (-kAe^{-kt})(1-kt) + (Ae^{-kt})(-k)$$

$$G''(t) = -kAe^{-kt} + k^2Ate^{-kt} - kAe^{-kt}$$

$$G''(t) = -2kAe^{-kt} + k^2te^{-kt}$$

$$G''(t)= -kAe^{-kt}(2 - kt)$$

We set the second derivative to 0 to find when glucose is decreasing the fastest.

$$0 = 2 - kt$$

Subtract both sides by 2 and divide by k to determine the value of t.

$$t = 2 / k$$

Plugging in the value we found for k, which is 0.0100 mg/dL per minute, we will get

$$t = 2 / 0.0100 \text{ mg/dL per minute}$$

$$t = 200 \text{ minutes}$$

The value of t tells us the inflection point of the first derivative, telling us that the concavity is concave up because of its positive value, and where the minimum value is, which in this case is 200 minutes. This is when insulin is most effective in this subject's body. Now, we can do the calculation of

Fastest glucose drop rate – peak glucose level = the time before a meal insulin should be injected

200 minutes – 100 minutes = 100 minutes

Discussion and Conclusion

Given the data provided by the graph, we can conclude that the best time to inject insulin so that its peak effectiveness will align with when glucose spikes is at 100 minutes prior to eating a meal. We did so by using the equation $G(t) = A \cdot t \cdot e^{-kt}$ and getting its derivative to find the peak glucose level, then getting its second derivative to find the rate at which glucose drops the fastest. These results reveal that the application of calculus to a glucose concentration curve is beneficial as it allows for the alignment of the peak glucose level to match up with the most effective time insulin will be working, so that normal glucose levels can be achieved after a person with diabetes eats a meal.

Through this project, I learned how calculus can be applied in real-world examples, more specifically to endocrinology and the optimal time for injection of insulin in people with diabetes to prevent harmfully high blood glucose levels. Calculus helps to find specific points in time of when glucose measurements are higher than average, making it easier to determine when the hormone insulin needs to work most efficiently. Furthermore, calculus is indispensable in healthcare fields and will always be beneficial in improving health for people in their everyday lives.

Works Cited

Kowalczyk-Korcz, Emilia, et al. “Super Bolus-a Remedy for a High Glycemic Index Meal in Children with Type 1 Diabetes on Insulin Pump Therapy?-A Randomized, Double-Blind, Controlled Trial.” *MDPI*, Multidisciplinary Digital Publishing Institute, 16 Jan. 2024, www.mdpi.com/2072-6643/16/2/263.

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